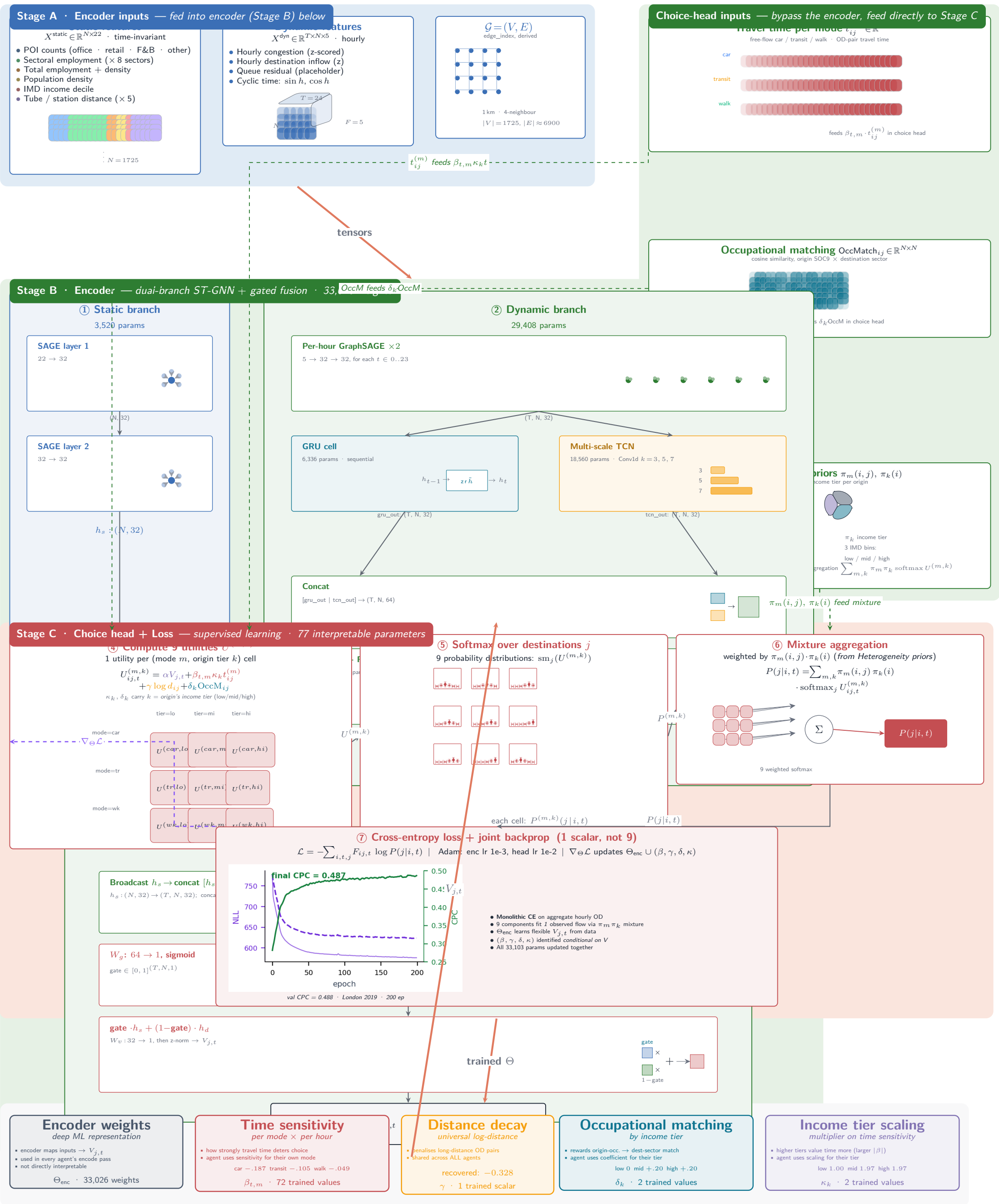


Figure 1 — ML Training Pipeline
supervised inverse learning of choice parameters from aggregate hourly OD



Trained Θ has 5 functional categories: Θ_{enc} (deep ML, used by anyone going through encoder); $\beta_{t,m}$ (time sensitivity, fetched per agent's mode); γ (distance, shared); δ_k (occ. match, fetched per tier); κ_k (tier scaling on β , fetched per tier). These five types bridge to Figure 2 (ABM forward simulation).